Name #1: ____________________

Name #2: ____________________

In-class Quiz
#02: Registers and Directives

Directives

1. (5 pts) The directive `Thumb` instructs the assembler to interpret subsequent instructions as 16-bit Thumb-1 instructions. If we want to interpret subsequent instructions as 32-bit Thumb-2 instructions, we should use `Thumb-2`.
   a. True
   b. False

2. (5 pts) Write the directive line to reserve an area in memory named `myCode`. The area should start at an address that is a multiple of 4 and will store your code.

3. (5 pts) Create a symbol named `myMain` to label the address of the entry point of your code, and let the assembler know the address.

Loading registers (move immediate numbers into registers

4. (5 pts) Write the single assembly instruction to load register **R2** with immediate hex value 0x32F.

   ANS: ______________________________________

5. (5 pts) Write the single assembly instruction to load register **R1** with the value currently stored in register **R0**.

   ANS: ____

Arithmetic operations

6. (5 pts) Write the single instruction to add the immediate hexadecimal value 2_11010101 to the value in register **R3** and store the result in register **R0**.

   ANS: ____

7. (5 pts) Write the single instruction to subtract the immediate decimal value 0x3E from the values currently in **R2** and store the result in register **R1**.

   ANS: ____

8. (5 pts) Write the single instruction to subtract the values currently in register **R1** from the values currently in **R3** and store the result in register **R3**.

   ANS: ____

9. (5 pts) Write the single instruction to multiply the value currently in register **R5** by the value currently in register **R6** and store the result in register **R4**.

   ANS: ____

10. (5 pts) Write the single instruction to divide the value stored in register **R1** by the value currently in register **R3** and store the result in to register **R0**. (unsigned division)

    ANS: ____

Programming (Can use multiple instructions)

11. (5 pts) Use register **R1** and **R2** to calculate the expression (3×5), and store the result in the register **R0.**

12. (5 pts) Use register **R1** and **R2** to calculate the expression $(15 \div 3)$, and store the result in the register **R0.** (unsigned division)

13. (10 pts) Use only one register **R0** to calculate the expression $(8 + 12 \times 12)$, and store the result in the register **R0.**

14. (10 pts) Use two register **R0** and **R1** to calculate the expression $(7 + 5 \times 9)$, and store the result in register **R0**. (unsigned division).

15. (10 pts) Use two register **R0** and **R1** to calculate the expression $(3 \times 6)/(3 + 6)$, and store the result in register **R0**. (unsigned division).

16. (10 pts) Use three register **R0**, **R1** and **R2** to calculate the expression $\frac{2\times3-4\times5-1}{5}$, and store the result in register **R0**. (signed division).

17. (Extra 10 pts) Use only one register **R0** to calculate the expression $(y - x^2)$, and store the result in register **R0**. Here, y and x can be any small values defined using EQU. (Hint: you can use MVN instruction.) Complete the following code:

```
x     EQU   10
y     EQU   3

      MOV   R0,#x
```